SAPIENZA
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Department of Mechanical and Aerospace Engineering

Homework 1 — Indirect Optimization of a Planar Ascent Trajectory

Dynamics and Control of Launch Vehicles

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Contents

1 Introduction

This report presents the solution to Homework 1 of the course *Dynamics and Control of Launch Vehicles*, which concerns the minimum-fuel planar ascent of a rocket from rest on the ground to a prescribed low-Earth orbit injection state. Four progressively richer mission programs are addressed, each framed as an optimal control problem and solved by applying the Pontryagin Maximum Principle (PMP) and single-shooting over the costate initial conditions.

1.1 Vehicle Dynamics

The vehicle is modelled as a point mass moving in a vertical plane above a flat, non-rotating Earth with no atmospheric drag. The state vector is

$$\mathbf{x}(t) = \left[x,\ y,\ v_x,\ v_y,\ m\right]^T, \tag{1}$$

where (x, y) is the position in the orbital plane, (v_x, v_y) is the velocity, and m is the vehicle mass. The single control variable is the thrust-direction angle φ measured from the horizontal. The thrust magnitude is $T = cQ$, where c is the effective exhaust velocity and $Q = -\dot{m}$ is the constant mass-flow rate. The equations of motion are

$$\dot{x} = v_x, \tag{2}$$

$$\dot{y} = v_y, \tag{3}$$

$$\dot{v}_x = \frac{T}{m}\cos\varphi, \tag{4}$$

$$\dot{v}_y = \frac{T}{m}\sin\varphi - g, \tag{5}$$

$$\dot{m} = -Q. \tag{6}$$

1.2 Non-dimensionalization

All quantities are made dimensionless with the reference values in Table 1. The reference velocity V_{ref} is the circular orbital speed at the nominal target altitude; the reference length and time follow from g.

Table 1: Non-dimensionalization reference values.

Quantity	Symbol	Expression	Value
Velocity	V_{ref}	orbital speed	7800 m/s
Acceleration	g_{ref}	g	9.81 m/s^2
Length	L_{ref}	V_{ref}^2/g	$\approx$ 6200 km
Time	t_{ref}	V_{ref}/g	$\approx$ 795 s
Mass	m_{ref}	m_0	10^6 kg

In non-dimensional form ($g = 1$, $m_0 = 1$, $v_{x,f} = 1$) the fixed problem data are collected in Table 2.

Table 2: Fixed non-dimensional problem parameters.

Symbol	Value	Physical interpretation
c	0.6	Effective exhaust velocity ($c_{\text{dim}} \approx 4680$ m/s, $I_{sp} \approx 477$ s)
η	0.1	Structural coefficient m_s/m_p
y_f	0.04, 0.05, 0.06	Target orbit altitude ($\approx$ 248, 310, 372 km)

The mass-flow rate Q is a free design parameter swept in Tasks 1 and 4. Since $W = mg$ and $m_0 = 1$, the initial thrust-to-weight ratio is $T/W\big|_{t=0} = cQ$, so liftoff ($T > W$) requires $Q > 1/c \approx 1.67$. As propellant burns off ($\dot{m} = -Q < 0$) the ratio $T/W = cQ/m$ increases monotonically, hence the liftoff instant is the binding case.

The value $Q = 2$ is the nominal mass-flow rate *given in the assignment data*. It corresponds to a liftoff thrust-to-weight ratio $T/W\big|_{t=0} = cQ = 1.2$—representative of a real first stage and safely above the $Q_{\min} = 1/c \approx 1.67$ liftoff threshold. Task 1 sweeps Q to study its effect and finds the payload-optimal rate $Q^* \approx 2.2$–2.5 (Table 3), so the nominal value sits just below the single-stage optimum. Tasks 2–4 then keep $Q = 2$ fixed, so that the comparison of the structural features (vertical climb, coast, staging) is made at a single, common operating point.

1.3 Pontryagin Maximum Principle

1.3.1 Hamiltonian and Costate Equations

The optimal control problem is to find the control $\varphi(t)$ and the free terminal time t_f that maximize the final mass $m(t_f)$—equivalently, minimize the Mayer cost $J = -m(t_f)$—subject to the dynamics (2)–(6) and the terminal constraints

$$y(t_f) = y_f, \qquad v_x(t_f) = 1, \qquad v_y(t_f) = 0. \tag{7}$$

The horizontal position $x(t_f)$ is free, and t_f is free.

Introducing the costate (adjoint) vector $\boldsymbol{\lambda} = [\lambda_x,\ \lambda_y,\ \lambda_{v_x},\ \lambda_{v_y},\ \lambda_m]^T$, the Hamiltonian is

$$\mathcal{H} = \lambda_x v_x + \lambda_y v_y + \lambda_{v_x}\frac{T}{m}\cos\varphi + \lambda_{v_y}\left(\frac{T}{m}\sin\varphi - 1\right) - \lambda_m Q. \tag{8}$$

Rearranging,

$$\mathcal{H} = \underbrace{\lambda_x v_x + \lambda_y v_y - \lambda_{v_y} - \lambda_m Q}_{\text{state-only part}} + \frac{T}{m}\underbrace{\left(\lambda_{v_x}\cos\varphi + \lambda_{v_y}\sin\varphi\right)}_{\text{control-dependent part}}. \tag{9}$$

The costate equations $\dot{\boldsymbol{\lambda}} = -\partial\mathcal{H}/\partial\mathbf{x}$ yield

$$\begin{aligned}
\dot{\lambda}_x &= 0 && \Longrightarrow \quad \lambda_x = 0 \quad (x(t_f) \text{ free } \Rightarrow \lambda_x(t_f) = 0), && (10)\\
\dot{\lambda}_y &= 0 && \Longrightarrow \quad \lambda_y = \text{const}, && (11)\\
\dot{\lambda}_{v_x} &= -\lambda_x = 0 && \Longrightarrow \quad \lambda_{v_x} = \lambda_{v_x,0} = \text{const}, && (12)\\
\dot{\lambda}_{v_y} &= -\lambda_y && \Longrightarrow \quad \lambda_{v_y}(t) = \lambda_{v_y,0} - \lambda_y t, && (13)\\
\dot{\lambda}_m &= \frac{T}{m^2}\left\|\boldsymbol{\lambda}_v\right\|, && && (14)
\end{aligned}$$

where $\|\boldsymbol{\lambda}_v\| = \sqrt{\lambda_{v_x}^2 + \lambda_{v_y}^2}$.

Four of the five costate components are determined analytically: $\lambda_x = 0$ identically, λ_y and λ_{v_x} are constant, and λ_{v_y} varies linearly in time. Only λ_m requires numerical integration via Eq. (14). This analytical structure greatly reduces the cost of evaluating the shooting residual.

1.3.2 Optimal Control Law

By the PMP, the optimal control φ^* maximizes $\mathcal{H}$ pointwise. Since $T/m > 0$, this is equivalent to maximizing the control-dependent part of Eq. (8),

$$g(\varphi) = \lambda_{v_x}\cos\varphi + \lambda_{v_y}\sin\varphi, \tag{15}$$

over φ. The first-order stationarity condition reads

$$\frac{\partial\mathcal{H}}{\partial\varphi} = \frac{T}{m}\big(-\lambda_{v_x}\sin\varphi + \lambda_{v_y}\cos\varphi\big) = 0 \qquad \Longrightarrow \qquad \tan\varphi^* = \frac{\lambda_{v_y}(t)}{\lambda_{v_x}}. \tag{16}$$

This is the **bilinear-tangent (linear-tangent) law**: since λ_{v_y} is linear in t (Eq. 13) and λ_{v_x} is constant, $\tan\varphi^*$ varies linearly with time.

Geometric interpretation and branch selection. Equation (16) determines φ^* only up to an additive π: it admits two roots per period. Geometrically, λ_{v_x} and λ_{v_y} are the two legs of a right triangle whose hypotenuse is the norm $\|\boldsymbol{\lambda}_v\| = \sqrt{\lambda_{v_x}^2 + \lambda_{v_y}^2}$, so the two candidate directions are

$$\text{(I)}\quad \cos\varphi^* = \frac{\lambda_{v_x}}{\|\boldsymbol{\lambda}_v\|},\ \sin\varphi^* = \frac{\lambda_{v_y}}{\|\boldsymbol{\lambda}_v\|} \qquad \text{or} \qquad \text{(II)}\quad \cos\varphi^* = -\frac{\lambda_{v_x}}{\|\boldsymbol{\lambda}_v\|},\ \sin\varphi^* = -\frac{\lambda_{v_y}}{\|\boldsymbol{\lambda}_v\|}. \tag{17}$$

Substituting each branch into Eq. (15) discriminates between them:

$$g_{\mathrm{I}} = \frac{\lambda_{v_x}^2 + \lambda_{v_y}^2}{\|\boldsymbol{\lambda}_v\|} = \frac{\|\boldsymbol{\lambda}_v\|^2}{\|\boldsymbol{\lambda}_v\|} = +\|\boldsymbol{\lambda}_v\|, \qquad g_{\mathrm{II}} = -\|\boldsymbol{\lambda}_v\|. \tag{18}$$

Branch (I) yields the maximum and branch (II) the minimum, so the PMP selects branch (I): the optimal thrust direction is aligned with the velocity costate $\boldsymbol{\lambda}_v$ (primer vector), and the control-dependent term attains its largest value,

$$\lambda_{v_x}\cos\varphi^* + \lambda_{v_y}\sin\varphi^* = \|\boldsymbol{\lambda}_v\|. \tag{19}$$

Using this result, the optimal Hamiltonian simplifies to

$$\mathcal{H}^* = \lambda_y v_y + \frac{T}{m}\|\boldsymbol{\lambda}_v\| - \lambda_{v_y} - \lambda_m Q. \tag{20}$$

1.3.3 Transversality Conditions

The costate boundary conditions follow from the transversality (natural boundary) conditions of the PMP. Adjoining the terminal constraints $\boldsymbol{\chi}(\mathbf{x}(t_f)) = [\,y - y_f,\ v_x - 1,\ v_y\,]^T = \mathbf{0}$ with multipliers $\boldsymbol{\mu}$ and the dynamics with the costates $\boldsymbol{\lambda}$, the first variation of the terminal (Mayer) payoff $\phi = m(t_f)$ produces a boundary term at t_f that vanishes only if

$$\boldsymbol{\lambda}(t_f) = \frac{\partial \phi}{\partial \mathbf{x}_f} + \left(\frac{\partial \boldsymbol{\chi}}{\partial \mathbf{x}_f}\right)^T \boldsymbol{\mu}. \tag{21}$$

Each terminal state falls into one of three categories. The *constrained* states y, v_x, v_y leave their costates $\lambda_y(t_f), \lambda_{v_x}(t_f), \lambda_{v_y}(t_f)$ undetermined: these absorb the unknown multipliers $\boldsymbol{\mu}$ and are carried as shooting unknowns (Eq. 24). The horizontal position x is *free and absent from the payoff*, so $\partial\phi/\partial x = 0$ and Eq. (21) returns $\lambda_x(t_f) = 0$, consistent with $\lambda_x \equiv 0$ already obtained in Eq. (10). Finally, the terminal mass $m(t_f)$ is *free but present in the payoff*, with $\partial\phi/\partial m = 1$, which fixes the only nontrivial terminal costate:

$$\lambda_m(t_f) = \frac{\partial \phi}{\partial m(t_f)} = 1. \tag{22}$$

This condition carries a twofold meaning. First, $\lambda_m(t_f)$ is the sensitivity of the optimal payoff to the terminal mass (a unit increase in $m(t_f)$ improves the objective by one unit). Second, since the PMP determines $\boldsymbol{\lambda}$ only up to a positive scale factor—the Hamiltonian is homogeneous of degree one in $\boldsymbol{\lambda}$—Eq. (22) fixes the normalization of the costate vector.[1]

Because the terminal time t_f is free, the PMP supplies one further condition—the natural boundary condition for t_f:

$$\mathcal{H}(t_f) = 0. \tag{23}$$

The system is autonomous (no explicit time dependence), so the Hamiltonian is constant along an optimal trajectory; $\mathcal{H}(t_f) = 0$ therefore implies $\mathcal{H}(t) \equiv 0$ throughout the powered arc, consistent with the optimality conditions.

[1] Notation follows the course notes (Lecture 4): ϕ is the terminal (Mayer) cost, $\boldsymbol{\chi}$ the boundary-constraint vector and $\boldsymbol{\mu}$ its multipliers (in the notes Φ denotes instead the running cost in $\mathcal{H} = \Phi + \boldsymbol{\lambda}^\top \mathbf{f}$, absent in this pure-Mayer problem). The lecture writes the final-node rule with a global minus sign, $\boldsymbol{\lambda}(t_f) = -\big(\partial\phi/\partial\mathbf{x}_f + \boldsymbol{\mu}^\top \partial\boldsymbol{\chi}/\partial\mathbf{x}_f\big)$, arising from the augmented functional in the $(\dot{\mathbf{x}} - \mathbf{f})$ form; the equivalent $(\mathbf{f} - \dot{\mathbf{x}})$ form adopted here carries the opposite sign and gives $\lambda_m(t_f) = +1$, consistent with the maximization form of $\mathcal{H}$ and with $\dot{\lambda}_m = (T/m^2)\|\boldsymbol{\lambda}_v\| > 0$ along the arc. Both conventions yield the same physical solution; the $+1$ normalization is the one used in the shooting residual and in `ode_burn.m`.

1.4 Single-Shooting Method and Continuation

Substituting the bilinear-tangent law (16) into Eqs. (2)–(6), the optimal control problem reduces to a multi-point **Boundary Value Problem (BVP)**. The shooting unknowns common to all four tasks are

$$\boldsymbol{\zeta} = \left[\lambda_{v_x,0},\ \lambda_{v_y,0},\ \lambda_y,\ \lambda_{m,0},\ t_f\right]^T. \tag{24}$$

Given a trial $\boldsymbol{\zeta}$, the augmented state $[x, y, v_x, v_y, m, \lambda_m]^T$ is integrated forward with `ode45` (tolerances: RelTol $= 10^{-10}$, AbsTol $= 10^{-12}$), and the residual vector is evaluated at t_f. The system is solved with `fsolve` (function and step tolerances both 10^{-10}, up to 500 iterations).

Single-shooting is sensitive to the costate initial guess, which has no direct physical interpretation. The **continuation strategy** mitigates this: the BVP is first solved at a favorable operating point ($Q \approx 3$), and each subsequent value of Q is initialized with the converged solution from the neighboring point, sweeping both forward and backward across the range $Q \in [1.8, 7]$. This warm-start approach propagates the solution branch smoothly without manual retuning of the initial guess.

2 Task 1: Single Burn Arc

2.1 Problem Statement

The simplest mission profile consists of a single continuous powered arc starting from rest at the origin, $(x, y, v_x, v_y) = (0, 0, 0, 0)$ with $m(0) = 1$, and ending at the orbital injection state satisfying Eq. (7). The mass-flow rate Q is the primary free parameter swept over the range $Q \in [1.8, 7]$; values below $Q_{\min} = 1/c \approx 1.67$ would yield insufficient thrust at liftoff ($T/W\big|_{t=0} = cQ < 1$).

2.2 Boundary Value Problem Formulation

The five shooting unknowns $\boldsymbol{\zeta}$ (Eq. 24) determine both the costate initial conditions and the free flight time. The shooting residual is

$$\mathbf{r}_1(\boldsymbol{\zeta}) = \begin{bmatrix} y(t_f) - y_f \\ v_x(t_f) - 1 \\ v_y(t_f) \\ \lambda_m(t_f) - 1 \\ \mathcal{H}(t_f) \end{bmatrix} = \mathbf{0}, \tag{25}$$

where $\mathcal{H}(t_f)$ is computed via Eq. (20). The first three components enforce the terminal state constraints (7); the fourth is the transversality condition (22) on the mass costate; the fifth is the free-time condition (23).

2.3 Velocity-Loss Decomposition

The total velocity loss $W_{\text{tot}} = \Delta V_{\text{ideal}} - v_f$ compares the ideal Tsiolkovsky delta-V,

$$\Delta V_{\text{ideal}} = c \ln \frac{m_0}{m_f} = c \ln \frac{1}{m_f}, \tag{26}$$

against the achieved injection speed $v_f = 1$. It is decomposed as

$$W_d = \int_0^{t_f} \frac{T}{m} \big[1 - \cos(\varphi - \psi)\big] \, \mathrm{d}t \qquad \text{(thrust misalignment loss)}, \tag{27}$$

$$W_g = \int_0^{t_f} \sin\psi \, \mathrm{d}t \qquad \text{(gravity loss)}, \tag{28}$$

where $\psi = \arctan(v_y/v_x)$ is the flight-path angle. Misalignment loss W_d arises from the fraction of thrust not aligned with the velocity vector; gravity loss W_g is the work done against gravity projected along the trajectory.

2.4 Results

Figure 1 shows the final mass as a function of Q for the three target altitudes. Each curve exhibits a clear interior maximum Q^*: at low Q, the long burn time accumulates large gravity losses; at high Q, the vehicle climbs steeply and thrust-misalignment loss dominates.

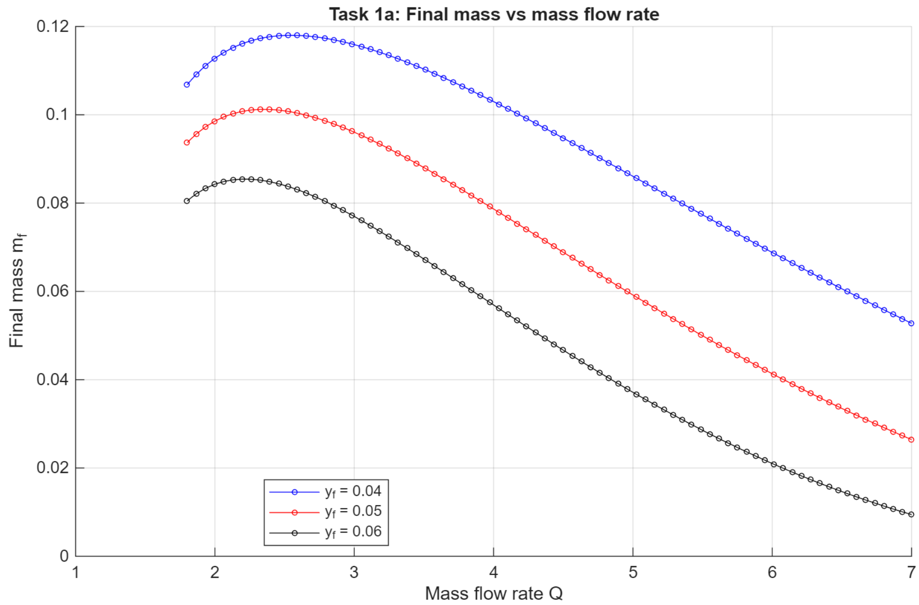


Figure 1: Task 1a: final mass m_f as a function of the mass-flow rate Q for three target altitudes y_f.

The interior maxima are quantified in Table 3: the optimal rate Q^* decreases from 2.52 to 2.19 as the target altitude rises from $y_f = 0.04$ to 0.06, and the delivered payload turns *negative* at $y_f = 0.06$—a single stage cannot reach that orbit with positive payload, which is precisely what motivates the staging of Task 4. The nominal $Q = 2$ of the assignment lies within a few percent of Q^*.

Table 3: Payload-optimal mass-flow rate Q^* (maximizing the final mass) for the three target altitudes, read from the Q-sweep of Figure 1. The delivered single-stage payload is $m_{\text{pay}} = m_f(1+\eta) - \eta$.

y_f	Q^*	m_f^*	m_{pay}^*
0.04	2.52	0.1180	+0.0298
0.05	2.33	0.1012	+0.0114
0.06	2.19	0.0854	−0.0060

Figure 2 confirms this trade-off by decomposing the velocity loss at $y_f = 0.04$. The gravity loss W_g is large at low Q and decreases monotonically with Q, while the misalignment loss W_d has a broad minimum near Q^*.

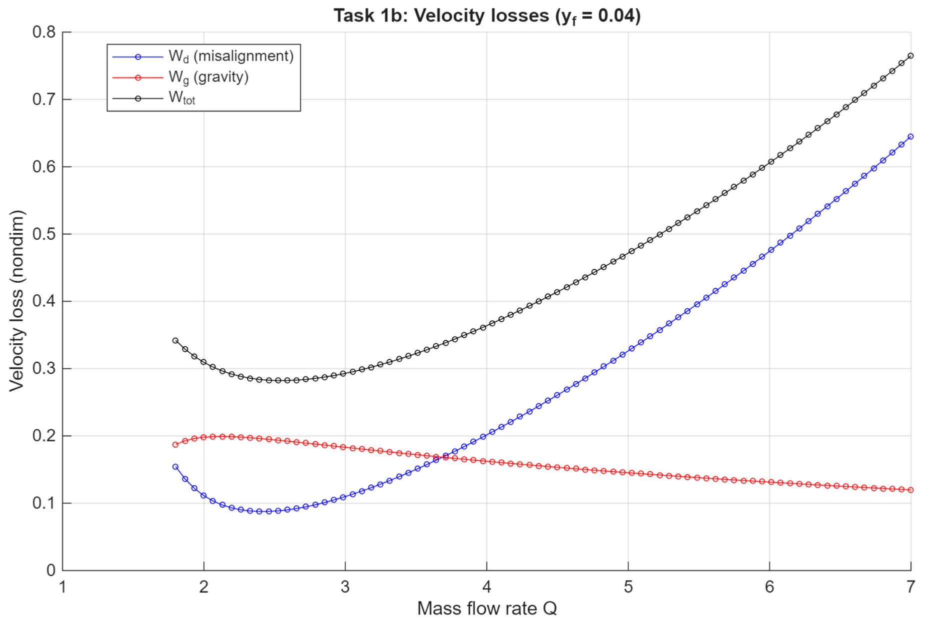
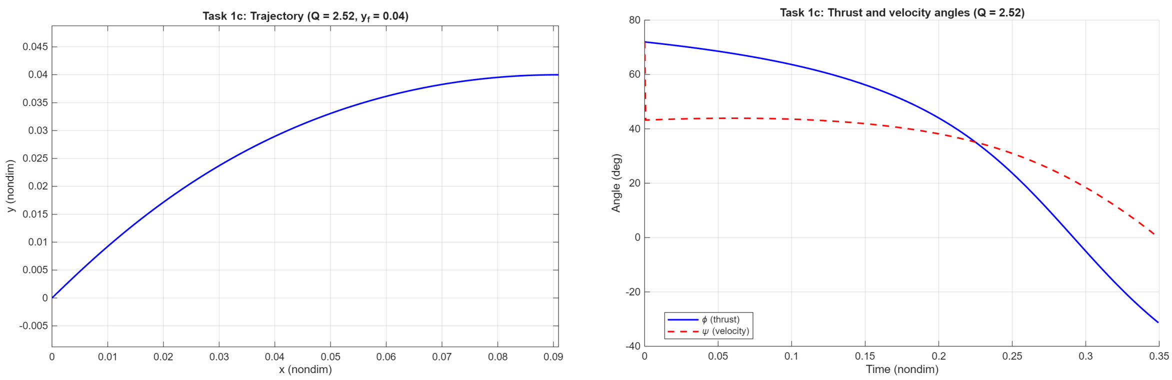


Figure 2: Task 1b: velocity-loss decomposition—misalignment W_d, gravity W_g, and total W_{tot}—at $y_f = 0.04$.

Figures 3a and 3b show the optimal trajectory and control history at $Q = Q^*$ for $y_f = 0.04$. The thrust angle φ decreases smoothly from near-vertical at launch to near-horizontal at injection, following the bilinear-tangent law (16). The velocity angle ψ tracks φ closely once the initial transient has died out.



(a) Optimal trajectory in the (x, y) plane.

(b) Thrust angle φ and velocity angle ψ vs. time.

Figure 3: Task 1c: optimal trajectory and control at $Q = Q^*$, $y_f = 0.04$.

3 Task 2: Vertical Climb and Optimal Burn

3.1 Phase 1: Vertical Climb

Practical launchers cannot pitch over immediately at liftoff because of aerodynamic loads and tower-clearance requirements. Task 2 prepends a short *vertical-climb phase* ($\varphi = \pi/2$, thrust pointing straight up) until the altitude reaches $y_1 = 10^{-4}$ (dimensionally ≈ 620 m). During this phase the horizontal position and velocity remain zero, and the 3-state sub-problem is

$$\dot{y} = v_y, \qquad \dot{v}_y = \frac{T}{m} - 1, \qquad \dot{m} = -Q, \tag{29}$$

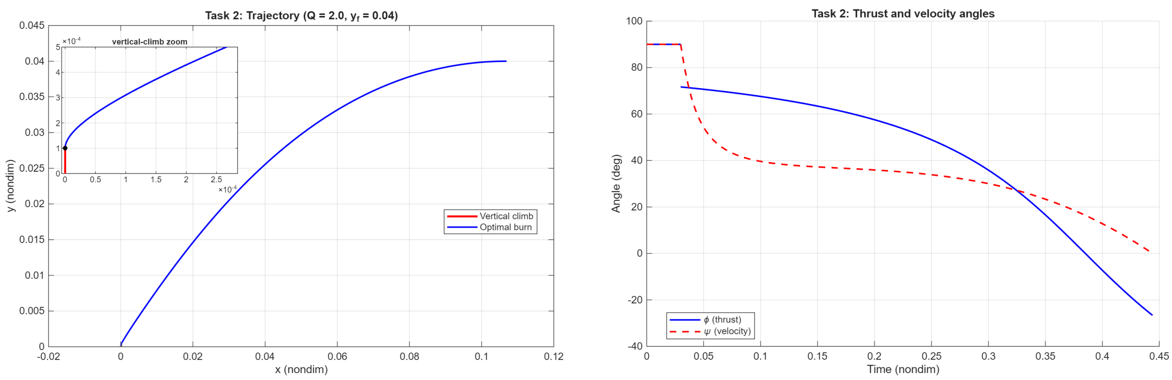
integrated until the event $y = y_1$ is detected via `ode45` event handling. The end state $(y_1,\ v_{y,1},\ m_1)$ becomes the initial condition for Phase 2.

3.2 Phase 2: Optimal Burn

Starting from the state $(x_0, y_0, v_{x,0}, v_{y,0}, m_0) = (0,\ y_1,\ 0,\ v_{y,1},\ m_1)$, Phase 2 is an optimal powered arc with the same BVP structure as Task 1. The shooting unknowns and the residual (25) are unchanged; only the initial conditions of the augmented state differ. Because the orbit injection targets are identical to Task 1, the continuation warm-start from Task 1's solution at $Q = 2$ provides a reliable initial guess.

3.3 Results

Figure 4 shows the composite trajectory and the angle history for $Q = 2$, $y_f = 0.04$. The red segment is the short vertical-climb phase; the blue arc is the optimal burn. The trajectory is qualitatively similar to Task 1 except for the initial vertical stub. During the climb both φ and ψ are fixed at 90°; at the transition the thrust begins to rotate toward the horizontal following the bilinear-tangent law, while the velocity angle ψ lags momentarily before catching up as horizontal speed builds.



(a) Composite trajectory: vertical climb (red), optimal burn (blue).

(b) Thrust angle φ and velocity angle ψ during the burn.

Figure 4: Task 2: trajectory and control history for $Q = 2$, $y_f = 0.04$.

The final mass in Task 2 is marginally lower than in Task 1: the vertical climb consumes propellant while building only vertical velocity, contributing to gravity losses without advancing the vehicle toward the horizontal injection speed. Nevertheless, the prepended climb is a standard launch practice, and the payload penalty is modest for the short y_1 value used here. The converged simulation results are collected in Table 4.

Table 4: Task 2 simulation results ($Q = 2$, $y_f = 0.04$). All quantities are non-dimensional; the Task 1 final mass at the same Q is reported for reference.

Quantity	Symbol	Value
Vertical-climb duration	t_1	0.0298
Powered-burn duration	t_{burn}	0.4142
Total flight time	t_f	0.4440
Final mass	m_f	0.1119
Delivered payload	m_{pay}	0.0231
Final mass — Task 1 (no climb)	$m_f^{(1)}$	0.1128

4 Task 3: Vertical Climb, Burn, and Coast

4.1 Switching Function and Thrust-Arc Structure

Task 3 introduces a *coast arc* between the powered burn and orbital injection. Unlike Tasks 1 and 2—where the engine burns throughout at the fixed magnitude $T = cQ$—the engine may now shut down: the thrust magnitude becomes a bounded control $T \in [0, T_{\max}]$ (with $T_{\max} = cQ$), alongside the steering angle φ, and the mass flow is $\dot{m} = -T/c$. Collecting in the Hamiltonian the terms proportional to T gives

$$\mathcal{H} = \underbrace{\lambda_x v_x + \lambda_y v_y - \lambda_{v_y}}_{\mathcal{H}_0 \ \text{(independent of } T)} + T\left(\frac{\lambda_{v_x}\cos\varphi + \lambda_{v_y}\sin\varphi}{m} - \frac{\lambda_m}{c}\right), \tag{30}$$

where the mass term $-\lambda_m \dot{m} = -\lambda_m T/c$ now sits inside the T-bracket because T is no longer constant. Inserting the optimal steering of the bilinear-tangent law, which maximizes the bracket by aligning the thrust with the velocity costate so that $\lambda_{v_x}\cos\varphi + \lambda_{v_y}\sin\varphi = \|\boldsymbol{\lambda}_v\|$ (Eq. 19), the Hamiltonian becomes *linear* in the thrust magnitude:

$$\mathcal{H} = \mathcal{H}_0 + T\,S(t), \qquad S(t) = \frac{\|\boldsymbol{\lambda}_v(t)\|}{m(t)} - \frac{\lambda_m(t)}{c}. \tag{31}$$

The coefficient $S(t)$ of T is the **switching function**: the marginal change of the Hamiltonian per unit thrust, i.e. the propulsive benefit $\|\boldsymbol{\lambda}_v\|/m$ (alignment of thrust with the primer vector $\boldsymbol{\lambda}_v$) net of the propellant penalty λ_m/c. By Pontryagin's maximum principle the optimal T maximizes $\mathcal{H}$ pointwise over $[0, T_{\max}]$; since $\mathcal{H}$ is linear in T, the maximizer lies at a bound—the **bang–bang** law

$$T^* = \begin{cases} T_{\max}, & S > 0 \quad \text{(engine on)}, \\ 0, & S < 0 \quad \text{(coast)}. \end{cases} \tag{32}$$

A coast arc is therefore an interval where $S < 0$; there $T = 0$, so $\dot{m} = 0$ and the vehicle flies ballistically. Because $\dot{\lambda}_m = 0$ whenever $T = 0$ and the transversality condition (22) fixes $\lambda_m(t_f) = 1$, the mass costate stays constant at $\lambda_m \equiv 1$ along the coast; by continuity $\lambda_m(t_c) = 1$ at cutoff, so there the switching function reduces to $S(t_c) = \|\boldsymbol{\lambda}_v(t_c)\|/m(t_c) - 1/c$.

The three-phase mission sequence is:

1. **Vertical climb**: $\varphi = \pi/2$ until $y = y_1 = 10^{-4}$ (identical to Task 2),
2. **Powered burn**: bilinear-tangent steering until engine cutoff at time t_c,
3. **Coast arc**: ballistic flight from t_c to the injection point.

4.2 Engine Cutoff Conditions

The shooting unknowns remain $\boldsymbol{\zeta} = [\lambda_{v_x,0},\ \lambda_{v_y,0},\ \lambda_y,\ \lambda_{m,0},\ t_{\text{burn}}]^T$ (with time measured from the start of Phase 2). At engine cutoff the following five conditions must be satisfied simultaneously:

$$\mathbf{r}_3(\boldsymbol{\zeta}) = \begin{bmatrix} v_{x,c} - 1 \\ y_c + \frac{1}{2}v_{y,c}^2 - y_f \\ \lambda_m(t_c) - 1 \\ \lambda_{v_y}(t_c) - \lambda_y\, v_{y,c} \\ S(t_c) \end{bmatrix} = \mathbf{0}. \tag{33}$$

Conditions 1 and 2 encode the *ballistic-coast reachability*: since during the coast v_x is constant and v_y decreases linearly under gravity, the vehicle must already have the orbital horizontal speed ($v_{x,c} = 1$) at cutoff, and the apogee of the ballistic arc must coincide with the target altitude. Condition 3 is the standard mass-costate transversality (22). Condition 4 ensures $\lambda_{v_y}(t_f) = 0$ at the coast end: since λ_{v_y} evolves linearly throughout, $\lambda_{v_y}(t_f) = \lambda_{v_y}(t_c) - \lambda_y v_{y,c}$, and setting this to zero recovers condition 4. Condition 5 is the bang-bang switching condition: the engine cuts off exactly when $S = 0$.

4.3 Coast Arc

During the ballistic coast ($T = 0$) the trajectory is analytically integrable. Denoting the elapsed coast time by τ and the cutoff state by subscript c:

$$x(t_c + \tau) = x_c + v_{x,c}\,\tau, \qquad y(t_c + \tau) = y_c + v_{y,c}\,\tau - \tfrac{1}{2}\tau^2. \tag{34}$$

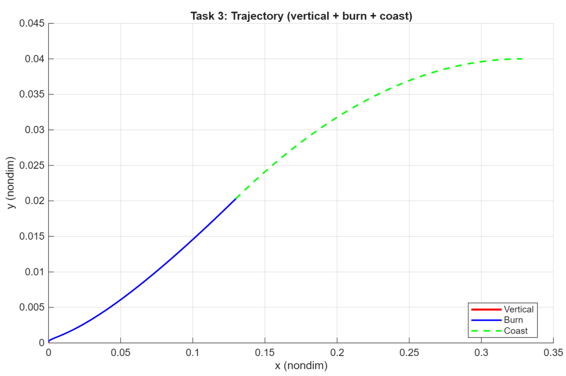
The coast ends at apogee $\tau^* = v_{y,c}$, where $v_y = 0$ and

$$y_f = y_c + \tfrac{1}{2}v_{y,c}^2, \tag{35}$$

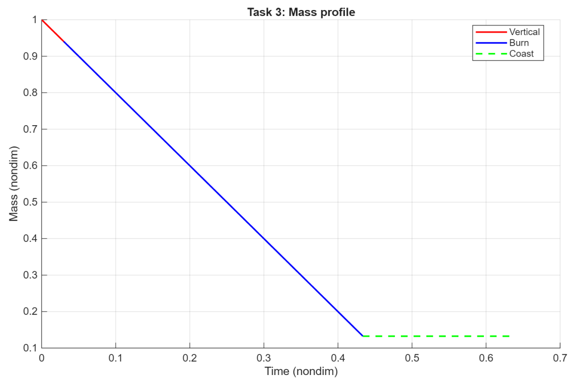
in agreement with condition 2 of (33). The vehicle mass is constant throughout the coast: $m_f = m_c$.

4.4 Results

Figure 5 shows the three-phase trajectory and mass profile for $Q = 2$, $y_f = 0.04$. The vertical climb (red) is short; the burn arc (blue) terminates below the target altitude; the ballistic coast (green dashed) ascends to the injection point. The mass profile confirms the model: it decreases linearly during both powered phases and remains constant during the coast. The final mass in Task 3 exceeds that of Task 2: by cutting the engine early and relying on the ballistic coast, less propellant is consumed while still satisfying all terminal constraints.



(a) Trajectory: vertical climb (red), burn (blue), coast (green dashed).



(b) Mass profile: linear decrease during powered phases, flat during coast.

Figure 5: Task 3: trajectory and mass profile for $Q = 2$, $y_f = 0.04$.

Figure 6 shows the thrust and velocity angles throughout the three phases. The dashed vertical lines mark the end of the vertical climb and the engine cutoff. During the coast no thrust angle is defined; the velocity angle ψ continues to evolve ballistically until it reaches zero at apogee.

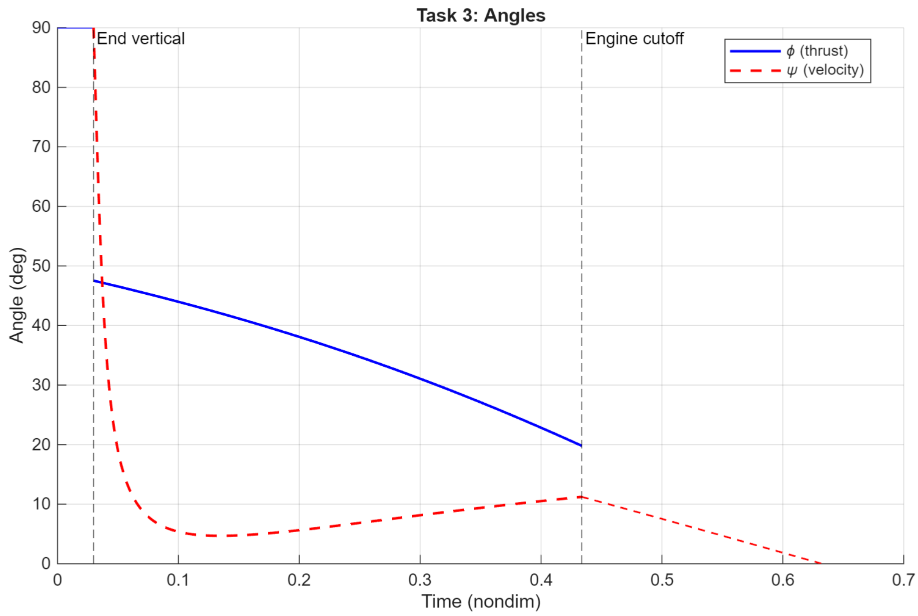


Figure 6: Task 3: thrust angle φ and velocity angle ψ across all three phases. Vertical dashed lines mark phase transitions.

A revealing contrast with Task 2 lies in the thrust angle just after the vertical climb: $\varphi \approx 47°$ here, against $\approx 71°$ in Task 2. Because the coast supplies the upper half of the climb (from $y_c \approx 0.020$ up to $y_f = 0.040$) with the engine off, the burn no longer has to gain all the altitude itself and can lean the thrust closer to the horizontal from the outset—a more aggressive gravity turn that spends a larger share of the thrust on horizontal acceleration. The same logic shows at cutoff: in Task 2 the thrust angle goes *negative* ($\varphi \approx -27°$), the engine pushing downward in the final instants to null

v_y at injection, whereas here it stays *positive* ($\varphi \approx +20°$)—the engine simply shuts off while v_y is still positive ($v_{y,c} = 0.199$) and lets gravity null it ballistically during the coast. Cutting off near-horizontally with residual vertical velocity, rather than burning propellant to decelerate vertically, is exactly what raises the final mass from $m_f = 0.112$ (Task 2) to 0.133.

The velocity angle ψ tells the same story from the trajectory's side. In Task 2 it falls from 90° to a plateau near 37° and then decreases monotonically to 0° *at injection*, with the engine still firing. In Task 3 it instead plunges to a much lower minimum ($\approx 5°$), then *rises* back to $\approx 11°$ at cutoff, and only reaches 0° *at apogee*, with the engine off. The deep minimum is the counterpart of the shallower thrust angle: with $\varphi \approx 47°$ and the mass still high just after the climb, the vertical thrust component $\frac{T}{m} \sin \varphi$ is smaller than g, so v_y drops and the path flattens sharply; as propellant burns off $\frac{T}{m}$ grows until $\frac{T}{m} \sin \varphi > g$ again, v_y climbs back and ψ recovers—which is precisely how the burn builds up the residual $v_{y,c} = 0.199$ handed to the coast. In Task 2 the larger φ keeps $\frac{T}{m} \sin \varphi > g$ from the start, so v_y never collapses, ψ holds its plateau, and the engine itself drives ψ to zero. The U-shaped ψ of Task 3, against the monotone ψ of Task 2, is thus the trajectory-side signature of the coast.

5 Task 4: Optimal Staging

5.1 Staging Model

Task 4 removes the vertical-climb phase and introduces a single *staging event* at time $t = t_s$. At staging, the first-stage structural mass

$$m_{s1} = \eta\, Q\, t_s \tag{36}$$

is jettisoned instantaneously. Here $Q\, t_s$ is the propellant the first stage burns up to t_s and $\eta = m_s/m_p$ is the *structural coefficient* (inert mass per unit propellant, $\eta = 0.1$): to hold that propellant the stage must carry a tankage/engine mass $\eta\, Q\, t_s$. The propellant is fully consumed—the mass falls as $m(t) = 1 - Qt$—and only the inert structure is dropped, so the vehicle mass undergoes a step:

$$m^+(t_s) = m^-(t_s) - m_{s1} = 1 - Q\, t_s - \eta\, Q\, t_s = 1 - Q\, t_s(1+\eta). \tag{37}$$

For $m^+(t_s) > 0$ one requires $t_s < [Q(1+\eta)]^{-1}$, which provides an upper bound on the admissible staging time. The costate vector is continuous across the staging event.

The net payload delivered to orbit is

$$m_{\text{pay}} = m_f - m_{s2}, \qquad m_{s2} = \eta\, Q\, (t_f - t_s), \tag{38}$$

where m_{s2} is the second-stage structural mass (again η times the propellant that stage burns). Summing the budget over both stages—a total propellant $Q\, t_f$, its structure $\eta\, Q\, t_f$, and the payload, all drawn from $m_0 = 1$—collapses to the compact relation

$$m_{\text{pay}} = 1 - (1+\eta)\, Q\, t_f, \tag{39}$$

which depends only on the *total* burn time t_f (equivalently, the total propellant). Staging raises the payload precisely because shedding the spent first-stage structure lets the vehicle reach orbit on a smaller total propellant load, i.e. a smaller t_f: here t_f drops from 0.444 (single stage) to 0.424 (two-stage), lifting m_{pay} from 0.024 to 0.068.

5.2 Optimization Procedure

For a fixed staging time t_s, the two burn arcs share the same costate evolution (the bilinear-tangent law is unaffected by staging), and the BVP reduces to the same structure as Task 1 with updated initial mass after staging:

1. Integrate the augmented state from $t = 0$ to $t = t_s$ (Stage 1 burn),
2. Apply the staging mass jump (36),
3. Integrate from t_s to t_f (Stage 2 burn),

4. Evaluate the residual (25) at t_f.

The shooting unknowns $\boldsymbol{\zeta}$ (Eq. 24) are identical to Task 1.

The optimal staging time t_s^* is found by an **outer parametric sweep**: t_s is varied over a dense grid of 50 points, the inner BVP is solved at each point (warm-started from the previous converged solution via continuation), and the payload (38) is maximized over the converged solutions. The starting guess for the inner BVP at the first grid point uses the single-stage solution from Task 1.

5.3 Results

Figure 7 shows the payload as a function of staging time t_s for $Q = 2$, $y_f = 0.04$. The curve has a well-defined maximum; the single-stage baseline (dashed red) lies significantly below the two-stage peak.

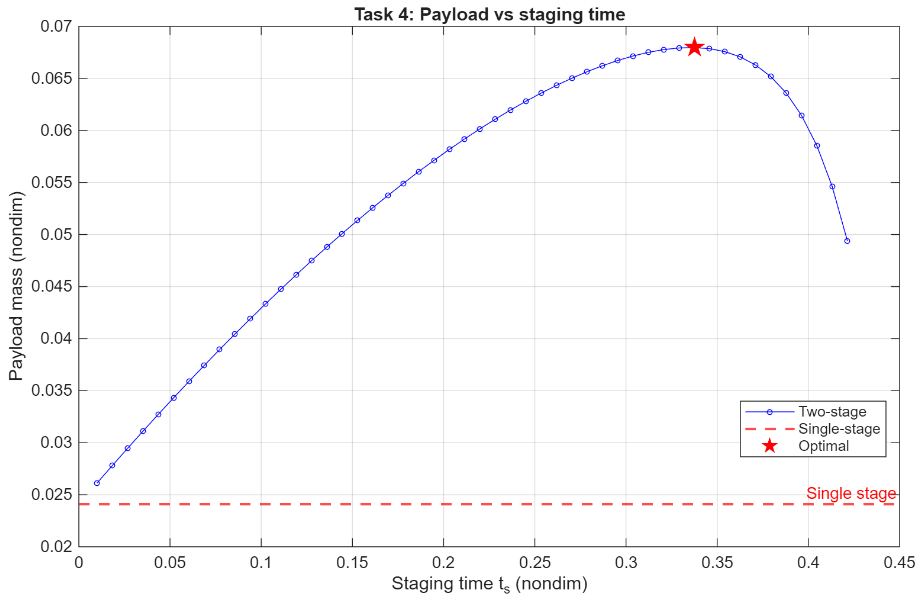


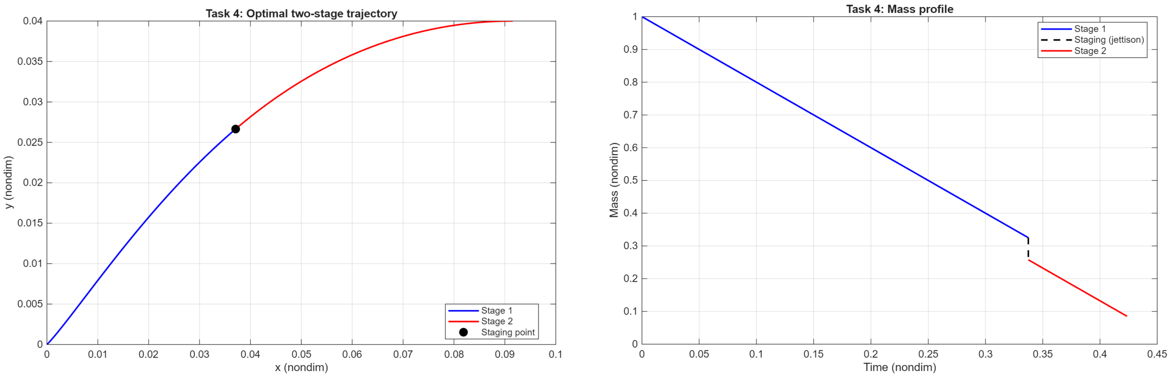
Figure 7: Task 4: payload mass vs. staging time t_s. Red dashed line: single-stage reference. Red star: optimal staging point.

Table 5 summarises the numerical results at $Q = 2$, $y_f = 0.04$.

Table 5: Task 4 key numerical results ($Q = 2$, $y_f = 0.04$).

Quantity	Value	Notes
Optimal staging time t_s^*	0.337	nondim ($\approx$ 268 s)
Single-stage payload	0.0241	nondim
Two-stage payload	0.0680	nondim
Payload gain	+182%	vs. single-stage

Figures 8a and 8b show the optimal two-stage trajectory and mass profile. The discontinuous mass drop at t_s^* is visible in the profile; the trajectory is continuous (position and velocity are unaffected by staging, only mass changes).



(a) Optimal two-stage trajectory: Stage 1 (blue), Stage 2 (red), staging point (circle).

(b) Mass profile with the instantaneous jettison at t_s^*.

Figure 8: Task 4: optimal two-stage solution for $Q = 2$, $y_f = 0.04$.

The large relative gain is a consequence of the model simplifications (constant Q, no drag, identical I_{sp} for both stages). Real launchers achieve much smaller but still substantial staging benefits; the key mechanism—jettisoning inert structural mass before it has to be accelerated to orbital velocity—is the same.

6 Conclusions

This work applied the Pontryagin Maximum Principle to four minimum-fuel planar ascent programs of increasing complexity, all solved by single-shooting with `fsolve` over the costate initial conditions. Table 6 summarises the four programs at the common operating point $Q = 2$, $y_f = 0.04$; the main findings are discussed below.

Table 6: Comparison of the four ascent programs at $Q = 2$, $y_f = 0.04$ (non-dimensional). The payload of the single-stage programs (Tasks 1–3) is $m_{\text{pay}} = m_f(1+\eta) - \eta$; for the two-stage program (Task 4) it is the net mass after jettisoning both stage structures.

Program	Mission phases	t_f	m_f	m_{pay}
Task 1	burn	0.444	0.1128	0.0241
Task 2	vertical + burn	0.444	0.1119	0.0231
Task 3	vertical + burn + coast	0.632	0.1326	0.0458
Task 4	stage 1 + stage 2 (staging)	0.424	0.0852	0.0680

The progression of delivered payload—0.0241 → 0.0231 → 0.0458 → 0.0680—quantifies the trade-offs: the vertical climb (Task 2) costs a few percent of payload relative to the bare burn (Task 1), the ballistic coast (Task 3) nearly doubles it (+90%), and optimal staging (Task 4) almost triples it (+182%). The main findings are as follows.

Bilinear-tangent law (all tasks). The flat-Earth, drag-free dynamics produce a particularly tractable adjoint structure: four of the five costate components are determined analytically (two constants and one linear ramp), leaving only the mass costate λ_m to be integrated numerically. The resulting bilinear-tangent steering law reduces the optimal control problem to a five-dimensional root-finding problem, making single-shooting computationally affordable.

Optimal thrust-to-weight ratio (Task 1). The mass-flow rate Q that maximises the final mass balances two competing losses: gravity losses (dominant at low Q, due to long burn times) and thrust-misalignment losses (dominant at high Q, due to steep climb angles). The optimal Q^* shifts slightly with target altitude y_f, reflecting the longer burn time required for higher orbits.

Continuation is indispensable. Single-shooting from a cold start fails across most of the Q range. The continuation strategy—warm-starting each new Q from the previous converged solution and sweeping forward and backward from a seed point—was the critical algorithmic ingredient enabling the parameter sweeps in Tasks 1 and 4.

Coast arc saves propellant (Task 3). Terminating the engine before reaching the target altitude and relying on ballistic dynamics is beneficial whenever the vehicle has sufficient residual vertical velocity to coast up to the injection point. The condition

$S(t_c) = 0$ determines the optimal engine cutoff time without an additional outer optimization loop.

Staging provides large payload gains (Task 4). At the reference operating point ($Q = 2$, $y_f = 0.04$), the optimal two-stage configuration delivers a +182% payload gain over the single-stage baseline. The fundamental mechanism—discarding first-stage structural mass before it needs to be accelerated to orbital velocity—is well-captured even by the simplified constant-Q, drag-free model.

Model limitations. The simplifications adopted throughout (flat Earth, no drag, constant thrust, no vehicle attitude dynamics) neglect key physical effects. Atmospheric drag dominates energy loss in the first 80 km of ascent; Earth's curvature and rotation alter the optimal coast and staging conditions; and real throttling constraints bound Q away from its theoretical range. Extending the framework to a drag model would shift Q^* and t_s^* quantitatively and, more fundamentally, would change the *structure* of the indirect problem. Because drag depends on altitude (through ρ) and on speed (through V), the costates $\lambda_y, \lambda_{v_x}, \lambda_{v_y}$ would no longer be constant or linear: the closed-form adjoints and the bilinear-tangent law would be lost, and all five costates would require numerical integration with a stiffer shooting problem. The optimal thrust direction would still follow the primer vector ($\tan\varphi = \lambda_{v_y}/\lambda_{v_x}$, since drag does not depend on φ), and the PMP plus single-shooting machinery would still apply—but the analytic structure that makes the present drag-free problem so tractable is precisely a consequence of having neglected aerodynamics.